Supplementary Material for QMOF-Rec

Representative RAG/LLM Interface Screenshots

The screenshots below illustrate representative query–response behavior in the implemented QMOF-Rec interface. They demonstrate interface behavior, grounded metadata reporting, missing-descriptor warnings, and refusal of unsupported validation claims. They are not part of the offline ranking validation.

Descriptor Availability and Masking

The reproducible QMOF-Rec evaluation uses 20,372 local vector-metadata records. Descriptor coverage was audited before revision: density is available for 20,372/20,372 records, band gap for 10,810/20,372 records, and void fraction for 0/20,372 records. Void fraction is therefore treated as unavailable rather than as zero, and porosity is excluded from active numerical ranking, diversity distances, and LEA candidate remapping.

Final Mask-Aware Rerun Artifacts

The final protocol was rerun with `configs/final_mask_aware_protocol.json`. The rerun uses five query scenarios, ten random seeds (42–51), top-$K = 5$, a main candidate-pool size of 100, and candidate-pool sensitivity at 50, 100, and 200. All ranking methods use the same saved candidate pool for a given query and seed. The complete machine-readable outputs are stored under `artifacts/final_masked_protocol/`, including candidate pools, per-method top-K identifiers, aggregate metrics, per-query/seed metrics, query-stratified summaries, variance decomposition, candidate-pool sensitivity, and WeightedSum versus LEA-no-diversity list-level comparisons.

Historical artifacts used in earlier drafts are preserved read-only under `artifacts/historical_protocol/`. Direct historical-versus-final comparisons are provided in `artifacts/protocol_comparison/`: `historical_vs_masked_metrics.csv`, `historical_vs_masked_topk.csv`, and `protocol_change_report.md`.

In the final rerun, WeightedSum, TOPSIS, ParetoCrowding, and baseline LEA tie on Rel@5 (0.8479) and NDCG@5 (1.0000) and have zero masked diversity under the active objective representation. MMR gives nonzero diversity (0.0400) with lower Rel@5 (0.8400), while graph-aware LEA variants preserve Rel@5 = 0.8479 and add small nonzero diversity (0.0070–0.0127). Candidate-pool sizes 50, 100, and 200 give identical aggregate LEA Rel@5 and NDCG@5 in this rerun, with runtime increasing as pool size increases.

Table 1: Descriptor coverage in the local metadata used for the revised evaluation.

Descriptor	Observed	Total	Coverage
Band gap	10,810	20,372	53.1%
Density	20,372	20,372	100.0%
Void fraction	0	20,372	0.0%

For each material m_i, QMOF-Rec stores an availability mask $\mathbf{a}_i$ in parallel with the normalized objective vector. Weighted relevance is computed only over observed objectives, and distances are computed only over jointly observed dimensions. This prevents a missing descriptor from being interpreted as a physical value of zero.

Retrieval Versus Reranking Missingness

The implemented retrieval layer and the numerical reranking layer use different representations. Retrieval uses SentenceTransformer embeddings of metadata text templates stored in a FAISS L2 index. The retrieval vector is therefore a text embedding rather than the numerical objective vector used by WeightedSum or LEA. In the revised vector-index builder, unavailable fields are written as unavailable metadata rather than as zero. The reproducible offline benchmark tables use a deterministic semantic proxy over metadata text and formula tokens, not a live sentence-transformer retrieval rerun.

Reranking uses active numerical objectives with masks. The active objectives are semantic score, band-gap score, density score, and stability score. Void fraction is excluded from relevance scoring, diversity distances, candidate remapping, objective balance, hypervolume proxy interpretation, and radar plots because it is unavailable for every local metadata record.

Detailed Computational Workflow

Algorithm 1 QMOF-Rec retrieval and objective scoring

Require: Query q, metadata corpus $\mathcal{D}$, QMOF records $\mathcal{M}$, candidate pool size n
Ensure: Candidate pool C_q, objective matrix X_q, weight vector $\mathbf{w}_q$

Embed query q and retrieve an initial semantic neighborhood from $\mathcal{D}$
Construct $C_q = \{m_1, \dots, m_n\}$ from the top retrieved material records
Infer or select query-dependent weights $\mathbf{w}_q$
for each material $m_i \in C_q$ **do**
    Compute semantic score $s_{\text{sem}}(m_i, q)$
    Compute available property scores, including band-gap and density scores where descriptors exist
    Build availability mask $\mathbf{a}_i$; do not impute missing descriptors as numerical zeros
    Exclude unsupported descriptors from active weighted scoring
    Form normalized objective vector $\mathbf{x}_i$
end for
Assemble X_q from all $\mathbf{x}_i$ and A_q from all $\mathbf{a}_i$
return $C_q, X_q, A_q, \mathbf{w}_q$

Algorithm 2 LEA-guided multi-objective reranking for QMOF-Rec

Require: Query q, candidate pool C_q, objective matrix X_q, availability matrix A_q, weight vector $\mathbf{w}_q$, population size N, iterations T, recommendation length K
Ensure: Top-K recommendation list S_K

Initialize population P^0 using candidate objective vectors and random vectors in $[0, 1]^d$
for $t = 1$ to T **do**
  for each individual $\mathbf{z}_i^t \in P^t$ **do**
    Map to nearest valid candidate $m_i = \pi(\mathbf{z}_i^t)$ using masked distance
    Compute $f(q, m_i, S)$ using the list-aware utility
    Generate mutant $\tilde{\mathbf{z}}_i^{t+1}$ using the implemented lotus mutation
    Map mutant to nearest valid candidate $\tilde{m}_i = \pi(\tilde{\mathbf{z}}_i^{t+1})$ using masked distance
    if $f(q, \tilde{m}_i, S) > f(q, m_i, S)$ **then**
      Accept $\tilde{\mathbf{z}}_i^{t+1}$
    else
      Retain $\mathbf{z}_i^t$
    end if
  end for
  if $t \bmod \tau = 0$ **then**
    Replace weakest ρN individuals by random normalized vectors
  end if
end for
Rank unique mapped QMOFs by LEA fitness and diversity-aware tie-breaking
return top-K materials S_K

Algorithm 3 Graph-aware RAG-assisted recommendation workflow

Require: Query q, QMOF graph $\mathcal{G}$, metadata corpus $\mathcal{D}$, material records $\mathcal{M}$
Ensure: Recommended materials with evidence-grounded explanation

Retrieve candidate pool C_q and evidence set $\mathcal{R}_K(q)$
Build or update material graph features $\mathbf{h}_i^{(0)}$
Encode QMOF graph with GraphSAGE or GAT when trained models are available
Compute optional graph similarity objective $s_{\text{graph}}(m_i, q)$
Run objective scoring and LEA-guided reranking to obtain S_K
Construct augmented prompt $A(q)$ from evidence, candidate records, scores, and uncertainty flags
Generate explanation y using the scientific assistant
Record feedback, logs, and analytics for future model improvement
return S_K, y

RAG/LLM Scoring Rubric and Hallucination Flags

Table 2: RAG/LLM evaluation dimensions and scoring rubric.

Dimension	Question	Scale	1 (Poor)	3 (Acceptable)	5 (Excellent)
Retrieval relevance	Does the RAG module retrieve relevant QMOF records for the query?	0–5	Irrelevant retrieval	Partially relevant	Directly useful for answering the query
Groundedness	Are the QMOF IDs and claims supported by retrieved candidates?	1–5	No retrieved IDs mentioned; context ignored	About half of IDs overlap with retrieved	All IDs drawn from retrieved candidates
Metadata consistency	Are QMOF IDs, formulas, density, and band-gap values consistent?	1–5	5+ inconsistencies	2 inconsistencies	No inconsistencies
Limitation awareness	Does the answer state when void fraction, porosity, or adsorption data are unavailable?	1–5	Makes unsupported adsorption/porosity claims	No explicit statement, but no false claims	Explicitly and correctly states missing descriptors
Explanation quality	Does the answer explain recommendation rationale?	1–5	Very short, no property discussion	Discusses some properties	Discusses properties, relevance, and uncertainty

Table 3: Binary hallucination flags used in the automated evaluation harness.

Flag	Meaning
Hallucination detected	Any of the flags below is true
Unsupported adsorption claim	Answer states or implies a specific CO_2 uptake, adsorption capacity, or gas-storage performance value
Unsupported porosity claim	Answer states or implies a specific or measured/confirmed porosity value
Experimental-validation claim	Answer claims a candidate is experimentally or laboratory validated
Graph misrepresentation	Answer describes GraphSAGE/GAT results as CIF-derived atomistic graphs rather than formula-derived metadata graphs
Metadata error	At least one mentioned QMOF ID is not in the retrieved set, or a stated formula does not match retrieved metadata
Missing-descriptor warning	Answer explicitly notes that a requested descriptor, such as void fraction or adsorption data, is unavailable

Repeated-Seed and Candidate-Pool Scope

The repeated-run artifacts use five query scenarios with ten random seeds per scenario. These 50 query/seed rows are useful for robustness checks, but they are nested within five query scenarios and should not be interpreted as 50 independent scientific queries. Candidate-pool-size sweeps are retained as supplementary reproducibility artifacts; they are used to examine retrieval–reranking sensitivity rather than to establish a universally optimal pool size.

Table 4: Query-stratified repeated-seed results for selected methods. Values are means and standard deviations over ten seeds within each query scenario. Seeds quantify stochastic algorithm variation within a fixed query, not independent scientific tasks.

Query	Method	Rel@5	SD	NDCG@5	SD	Div.	SD
q1_co2_adsorption	LEA baseline	0.4711	0.0005	0.9958	0.0014	0.0781	0.0072
q1_co2_adsorption	MMR	0.4654	0.0000	0.9856	0.0000	0.1211	0.0000
q1_co2_adsorption	WeightedSum	0.4727	0.0000	1.0000	0.0000	0.0086	0.0000
q2_photocatalysis	LEA baseline	0.8608	0.0001	0.9999	0.0001	0.0216	0.0014
q2_photocatalysis	MMR	0.8575	0.0000	0.9950	0.0000	0.0815	0.0000
q2_photocatalysis	WeightedSum	0.8608	0.0000	1.0000	0.0000	0.0211	0.0000
q3_lightweight_storage	LEA baseline	0.6657	0.0016	0.9861	0.0038	0.5687	0.0845
q3_lightweight_storage	MMR	0.6695	0.0000	0.9939	0.0000	0.5381	0.0000
q3_lightweight_storage	WeightedSum	0.6729	0.0000	1.0000	0.0000	0.0024	0.0000
q4_balanced_discovery	LEA baseline	0.5933	0.0000	1.0000	0.0000	0.0203	0.0002
q4_balanced_discovery	MMR	0.5808	0.0000	0.9768	0.0000	0.1115	0.0000
q4_balanced_discovery	WeightedSum	0.5933	0.0000	1.0000	0.0000	0.0202	0.0000
q5_insulating_frameworks	LEA baseline	0.8788	0.0006	0.9977	0.0008	0.1425	0.0243
q5_insulating_frameworks	MMR	0.8693	0.0000	0.9854	0.0000	0.2003	0.0000
q5_insulating_frameworks	WeightedSum	0.8808	0.0000	1.0000	0.0000	0.0319	0.0000

Table 5: Direct variance decomposition for mean relevance in repeated-seed artifacts. Between-query variance is much larger than within-query seed variance, supporting the interpretation that query scenario, not random seed, is the main dependence unit.

Method	Between-query var.	Within-query seed var.	Ratio
LEA baseline	0.030648	0.000001	49,331.3
MMR	0.030754	0.000000	$> 10^6$
RandomRepeated	0.030125	0.000005	6,215.4
SemanticOnly	0.030144	0.000000	$> 10^6$
WeightedSum	0.030560	0.000000	$> 10^6$

Table 6: Candidate-pool-size sensitivity for the available repeated-seed LEA pool variants. The saved artifacts contain pool sizes 50, 100, and 200. Pool size 100 is retained as the predefined main experimental setting to maintain consistency with the primary evaluation, not because this sensitivity analysis establishes it as optimal. Pool sizes 50–200 produce broadly similar quality, while runtime increases moderately with pool size.

Pool	Rel@5	NDCG@5	Diversity	Unique top-K frac.	Runtime (ms)
50	0.6945	0.9968	0.1739	1.000	123.65
100	0.6939	0.9959	0.1662	1.000	129.17
200	0.6936	0.9948	0.1716	1.000	143.40

Table 7: Metric-level check of LEA no-diversity versus WeightedSum in the saved repeated-seed artifacts. All 50 query/seed rows have identical saved Rel@5 and diversity values. The saved CSV does not include top-K identifiers, so top-K identity and ordering could not be verified from the available artifact.

Check	Rows	Result
Same Rel@5	50/50	yes
Same diversity	50/50	yes
Same top-K IDs	0/50	not available in saved metrics CSV
Same top-K order	0/50	not available in saved metrics CSV

Observed Physical Descriptor Means: All Records Versus Selected Top-K Portfolios

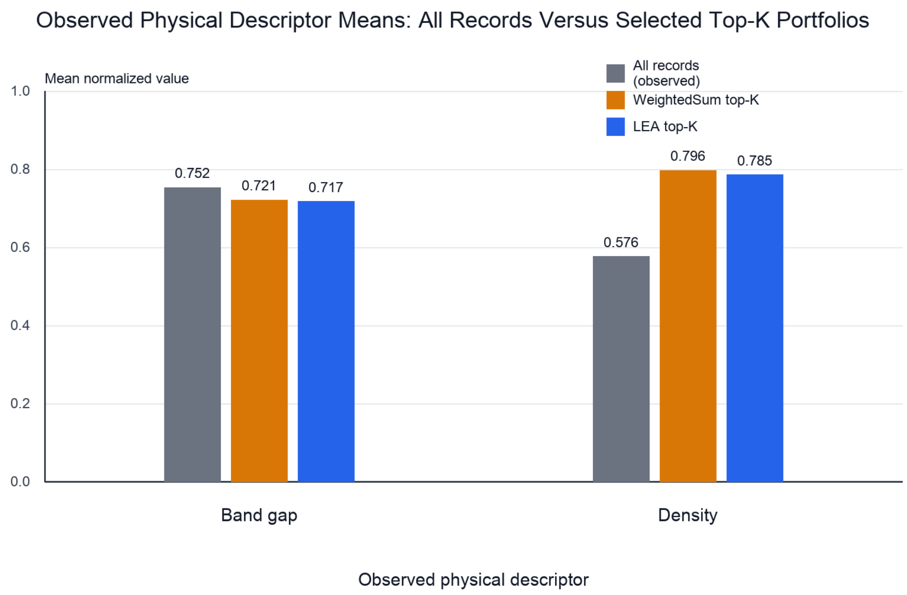


Figure S1: Observed physical descriptor means for all local records versus selected top-K portfolios. The figure includes only band gap and density because these are the observed physical descriptors available for comparison in the local metadata; semantic and stability scores are excluded from this physical-descriptor plot, and void fraction/porosity is excluded because void fraction is unavailable for all local records. Selected top-K values are averaged over the main saved single-run ranking artifact.

Missing-Data Ablation Status

A full missing-data ablation comparing zero filling, neutral filling, and masked scoring was not added to the numerical results because the saved repeated-seed artifacts do not include the ranked top-K identifiers needed to quantify changed top-K lists, and rerunning the full historical pipeline after the mask-aware code revision would require regenerating all primary and ablation tables under a new protocol. Instead, the repository now implements the preferred masked method and includes focused regression tests that verify missing descriptors are not treated as physical zero values.

CO_2 Adsorption Scenario

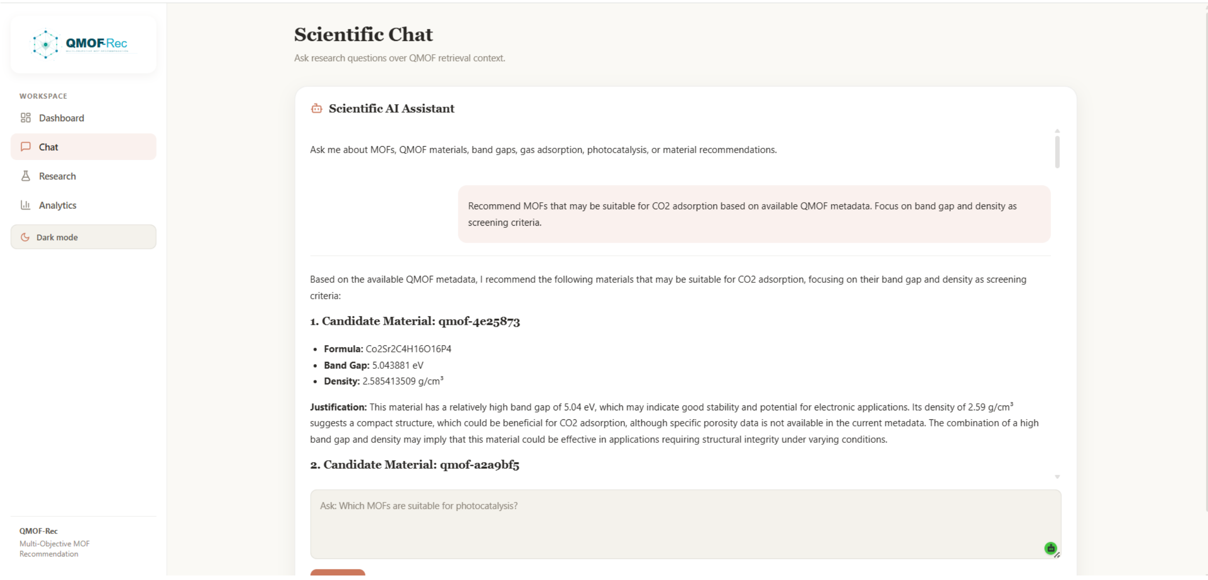


Figure S2: CO_2 adsorption scenario. The assistant recommends candidates based on available metadata and states that porosity and CO_2 uptake data are unavailable. Void fraction and adsorption uptake are unavailable in the current metadata.

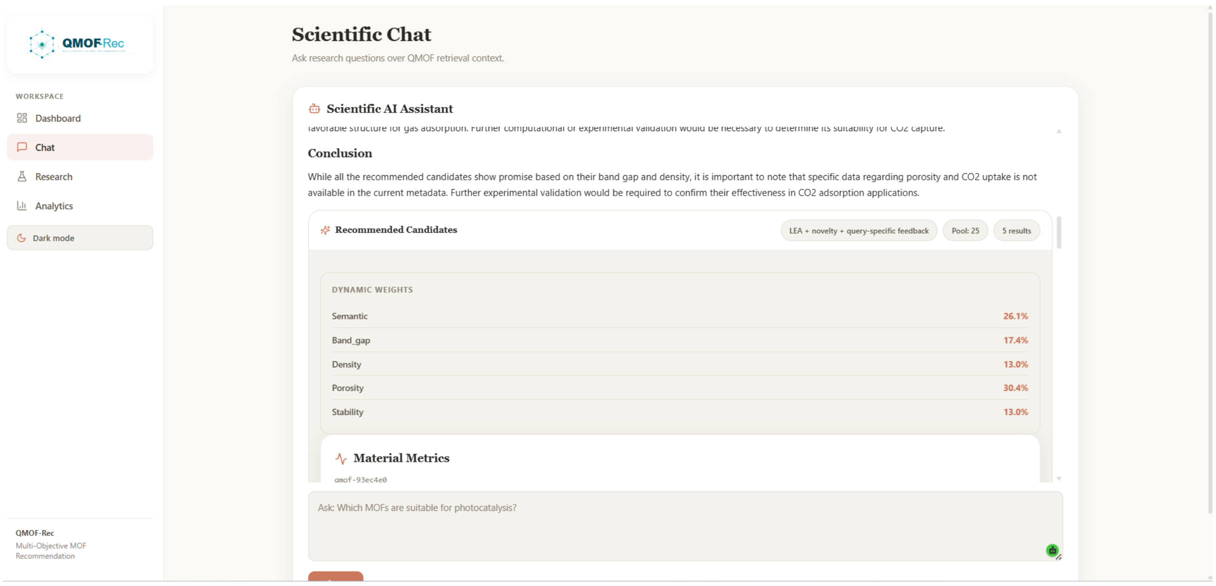


Figure S3: CO_2 adsorption scenario with structured candidate justifications and missing-descriptor warnings. Screenshots demonstrate the implemented interface and are not part of the offline ranking validation.

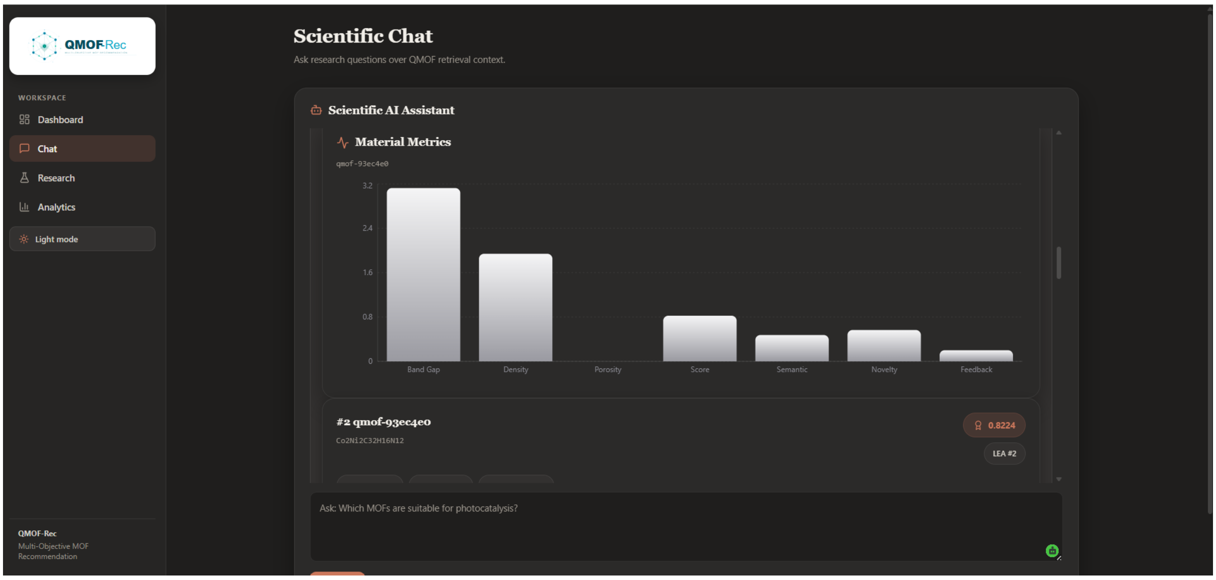


Figure S4: CO_2 adsorption scenario showing material metrics for a candidate record. In the revised implementation, void fraction is displayed only as unavailable metadata and is excluded from active numerical ranking.

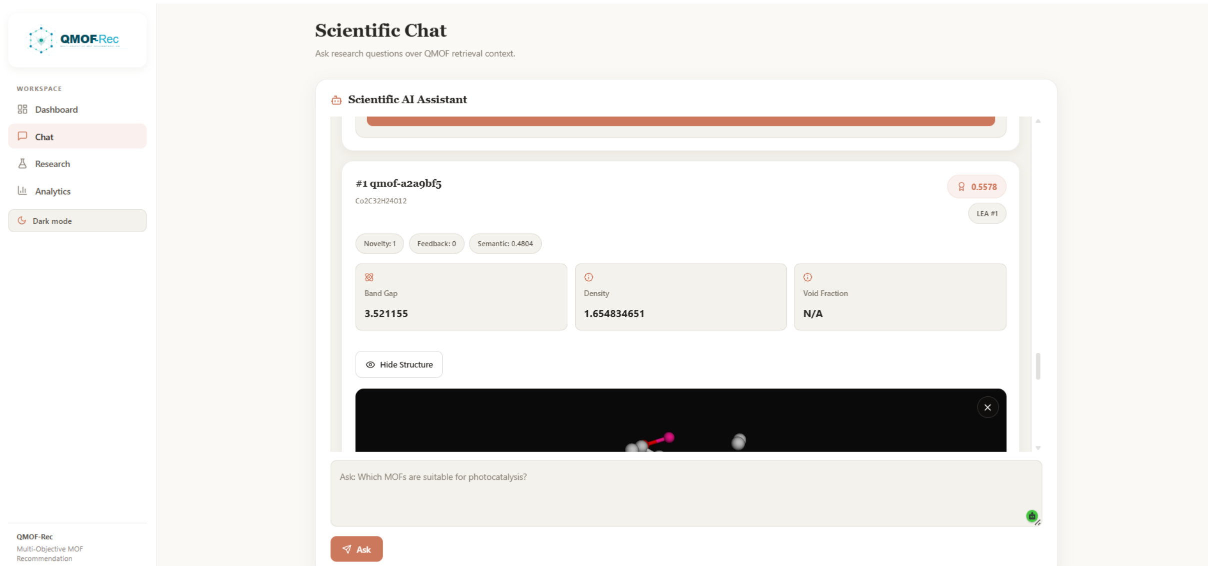


Figure S5: CO_2 adsorption scenario testing unsupported uptake claims. The assistant declines to identify experimentally validated CO_2 uptake because the retrieved context contains no experimentally validated uptake data.

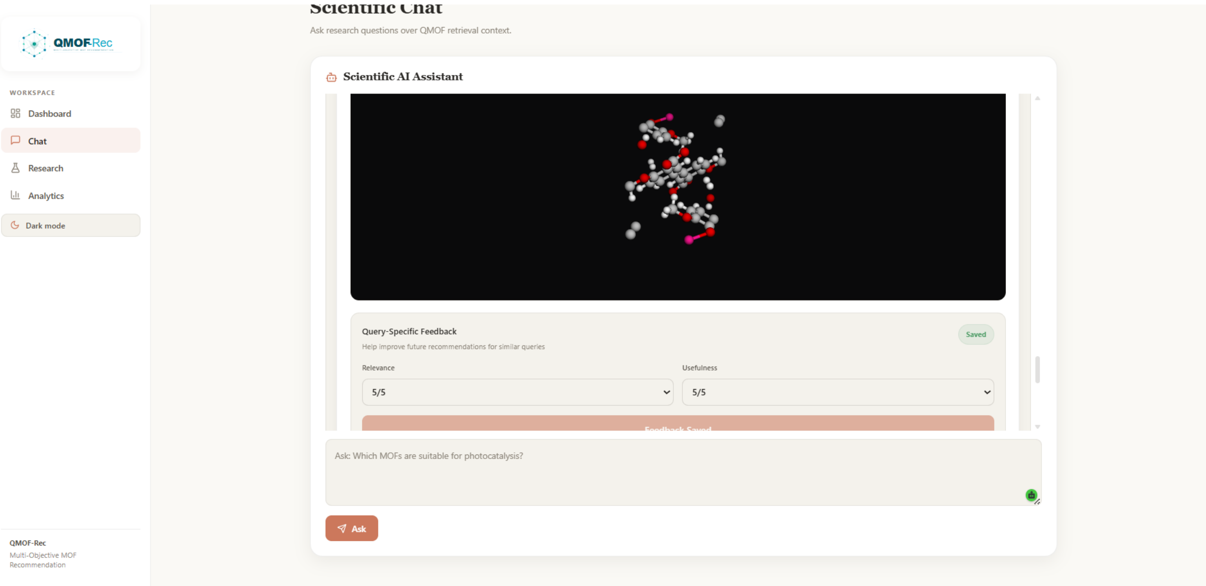
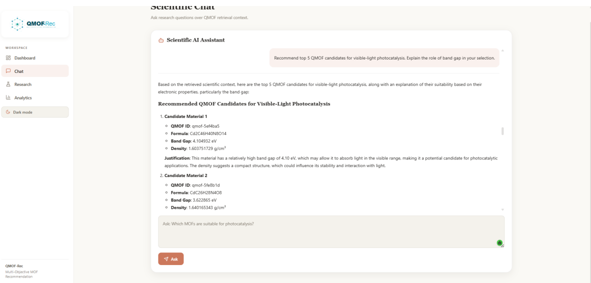
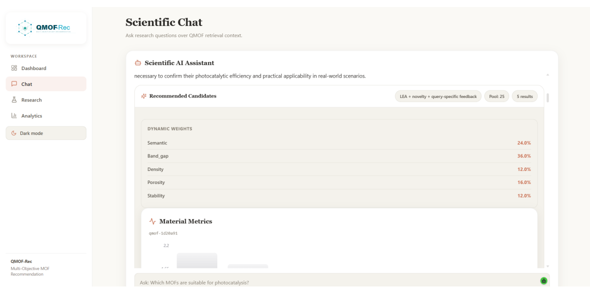


Figure S6: CO_2 adsorption scenario testing industrial-readiness claims. The assistant treats the listed MOFs as recommendation candidates requiring expert follow-up, not as experimentally validated capture materials.

Photocatalysis Scenario



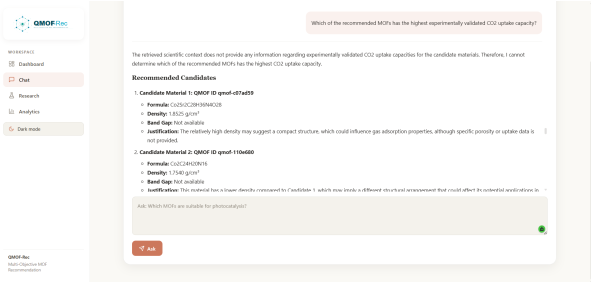
(a) Top-5 photocatalysis candidates with band-gap-oriented justifications.



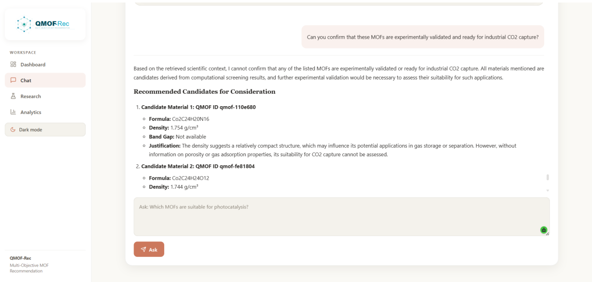
(b) Dynamic objective weights for a photocatalysis-oriented query.

Figure S7: Photocatalysis scenario screenshots. These illustrate answer structure and interface behavior rather than physical validation.

Missing-Information Trap



(a) Refusal to confirm experimentally validated CO_2 uptake.



(b) Refusal to confirm industrial readiness.

Figure S8: Missing-information trap screenshots. The assistant declines to fabricate experimental validation claims.

Lightweight Gas Storage

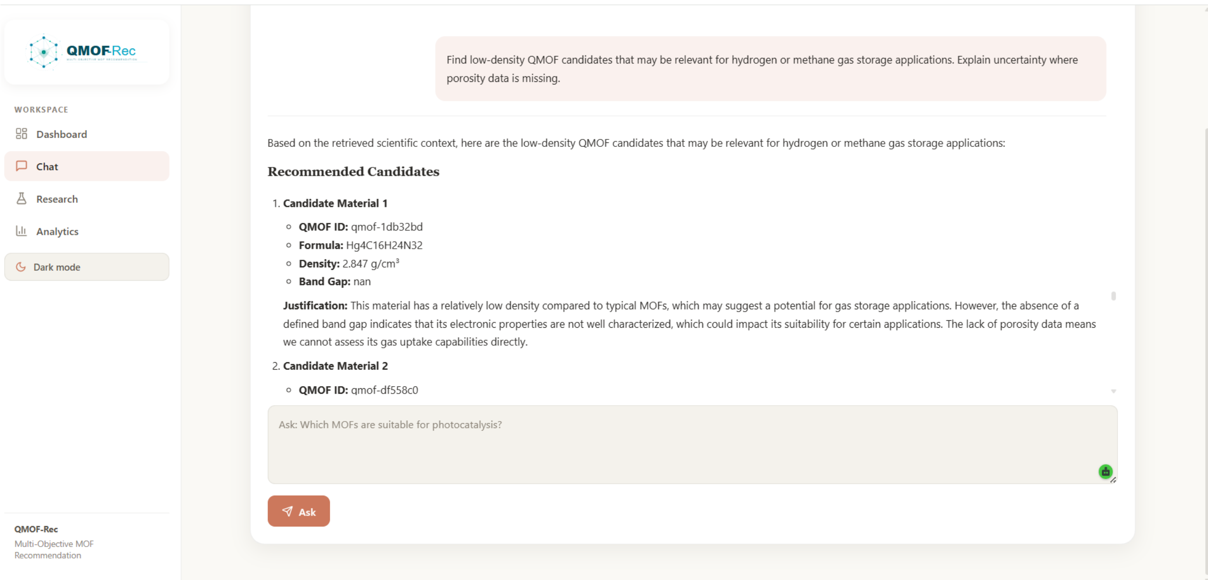


Figure S9: Lightweight gas-storage scenario. Candidate recommendations are based on available metadata such as density; void fraction, porosity, and gas uptake data are unavailable, so conclusions are preliminary.

Ranking Logic

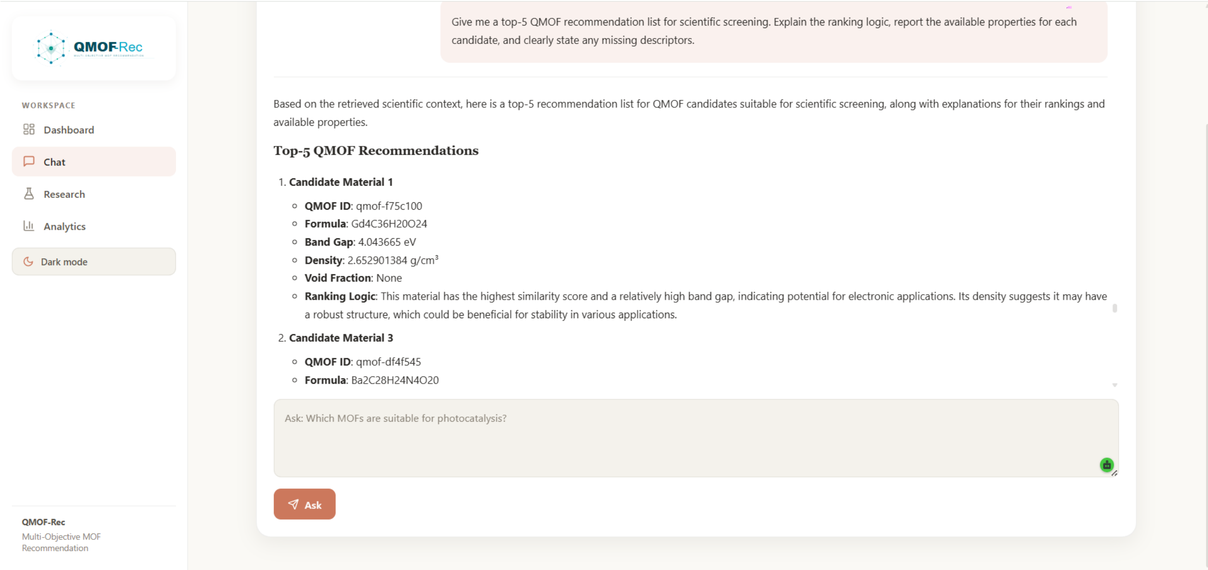


Figure S10: Ranking-logic scenario. The assistant provides a structured recommendation list with explicit ranking rationale and missing-descriptor reporting.

Chemically Novel Candidates

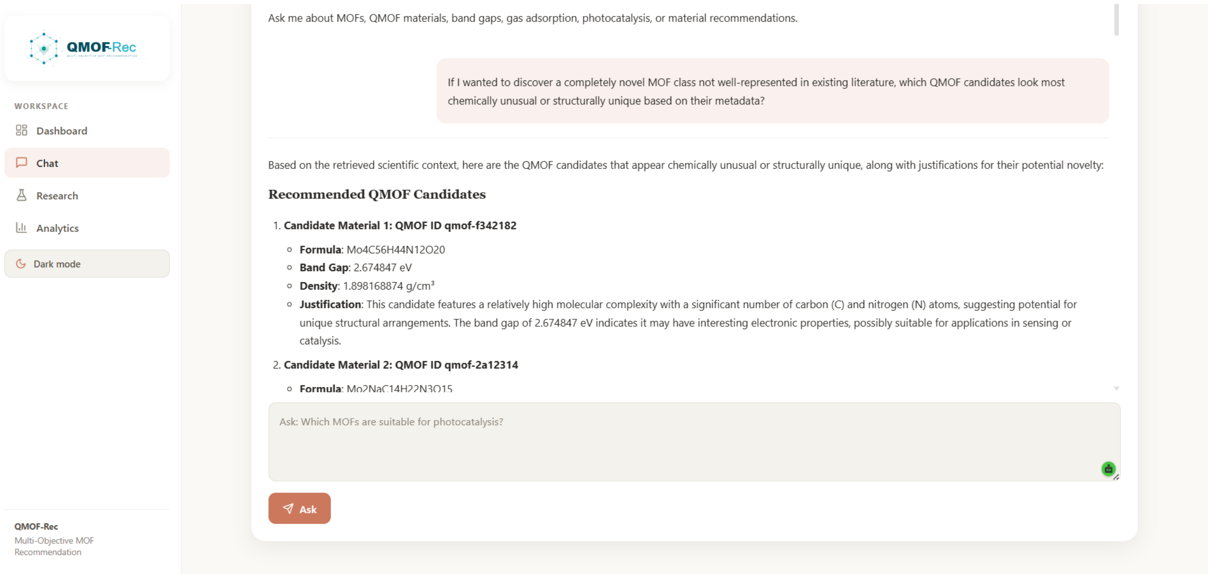


Figure S11: Chemically novel candidate scenario. The response is grounded in available metadata and should not be interpreted as experimental, adsorption, porosity, or structure-level validation.